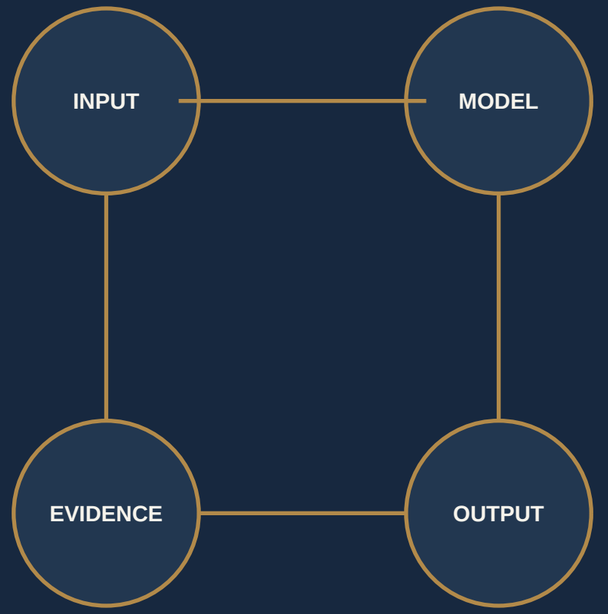
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Pattern Recognition

AI.211 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define statistical pattern recognition.
- Apply key methods including feature extraction.
- Build or analyze a practical system involving classification.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Statistical pattern recognition

02

METHODS

Feature extraction

03

SYSTEMS

Classification

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Statistical pattern recognitio

Statistical pattern recognition

02 · Feature extraction

Feature extraction

03 · Classification

Classification

04 · Clustering

Clustering

05 · Dimensionality reduction and

Dimensionality reduction and recognition systems

06 · Mandatory practical: Build rec

Mandatory practical: Build recognition systems using real-world data

Core ideas, made explicit

AI.211 • Distinguish definitions, assumptions and evidence

- Statistical pattern recognition
- Feature extraction
- Classification
- Clustering
- Dimensionality reduction and recognition systems

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.211 • Distinguish definitions, assumptions and evidence

- Mandatory practical: Build recognition systems using real-world data

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
Statistical pattern recognition

INPUTS
Feature extraction

METHOD
Classification

OUTPUT
Clustering

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving statistical pattern recognition.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using statistical pattern recognition.
- Implement or simulate a baseline using feature extraction.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.